LuminaGarden - Assignment 4: Alpha & Production Plan

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Alpha checkpoint: October 29-November 1. The complete garden loop, coherent wind, and day/night are integrated. Start empty, grow 1-5 plants, finish once all are mature, and restart. The existing seeded geometry, staged growth, colored point lights, and bloom remain the core.

Weekly development schedule

Calendar planned against a tentative December 14, 2026 submission date supplied by the team. Alpha checkpoint: October 29–November 1. Weekly plan: November 2–December 13. Confirm the official deadline if it changes; preserve the final buffer week.

Week / dates	Engineering tasks	Asset tasks	Integration / acceptance
1 / Nov 2–8	Stabilize the alpha loop and environment controls; profile on a second desktop GPU/browser; prototype basic shadows within budget.	Review existing soil, leaves, vein emission, and day/night palettes; correct readability and repetition defects.	Retest start, growth, wind, lighting, finish, and restart together. Keep or defer shadows based on measured visual benefit and FPS.
2 / Nov 9–15	Add a bounded movable light and controlled bending toward it; combine its deformation with wind without detaching organs.	Create a small procedural light marker; tune color/range presets and check plant/light readability.	Test simultaneous growth, wind, and light movement. Acceptance: repeatable seeds and five-plant limits still hold; lighting stays coherent.
3 / Nov 16–22	Add constrained Crystal Bloom, Spiral Vine, and Glowing Tree rule presets; connect a species selector to seed generation.	Build species-specific procedural leaves/petals and silhouettes; review palettes and record any new asset provenance.	Run every family through the same growth/wind/render pipeline. Test bounds, five-plant mixed gardens, and reset cleanup.
4 / Nov 23–29	Integrate species, wind, movable lighting, and shadows retained after profiling; exercise mixed five-plant gardens and long sessions.	Resolve inconsistent silhouettes, materials, light markers, and UI labels across species; audit asset provenance.	Produce a cross-feature test build. Acceptance: each family grows and sways coherently; no scene/light accumulation after repeated restart.
5 / Nov 30–Dec 6	Complete cross-feature integration and performance pass; resolve high-priority defects; freeze feature scope at the end of this week.	Finalize existing textures, palettes, and silhouettes; verify attribution and the consistency of all UI states.	Publish a release candidate on Vercel and test two desktop browsers. Feature freeze December 6; defer unstable optional work before the freeze.
6 / Dec 7–13	Bug fixes and regression checks only. Correct crashes, interaction defects, and performance regressions; no new features.	Polish existing asset/UI defects only; prepare final screenshots and documentation. No new feature assets.	Finalize README, PDF/release notes, and presentation/demo materials; verify hosted URL and restart flow. Submit on tentative December 14. No new features.

Kartik: procedural geometry, wind shaders, and profiling. Safa: UI/environment controls, asset review, and documentation. Both: integration, visual review, and regression testing.

Scope control: feature freeze December 6. December 7-13 is exclusively for fixes, documentation, and polish. Defer unstable optional features before the freeze; no new features or feature assets in the buffer.

Playable alpha and source

Playable project: https://lumina-garden-zeta.vercel.app/

Production alpha verified at this URL.

Public source / README: https://github.com/KartikP12s/Lumina-Garden

Run locally: npm ci; npm run dev. Production: npm run build; npm run preview. Node.js 22.12+ and desktop WebGL2 required. README contains build instructions.

Release notes - changes since Assignment 3

New features

- Complete loop: Start Garden menu → empty active garden → completion summary → restart/menu. Finishing requires at least one plant and no unfinished growth.

- Wind slider drives a shared plant-space shader bend after instance placement; stems, leaves, petals, and flower centers move coherently during and after growth.

- Day/night slider changes background, fog, and scene lighting. Emission and bloom soften in daylight to preserve plant readability.

- Menu/resume pauses the experience clock. Restart restores defaults and releases old plants, lights, materials, and instance buffers.

- HUD shows plant count, growth stage, seed hash, height, and branch/leaf counts. The five-plant limit remains enforced across manual planting and diagnostics.

- Benchmark exports alpha version, wind/daylight, effective bloom strength, browser/GPU, viewport, frame times, draw calls, and triangles. Vercel configuration and CI checks are included.

Bug fixes

- Camera drags that return to their starting point no longer count as planting clicks; pointer cancellation clears pending placement.

- Corrected daytime glow washout by attenuating emission and bloom as daylight increases.

- Guarded empty/unfinished completion and blocked conflicting controls during diagnostics. WebGL2 errors remain visible in the start menu.

Known issues and limits

- Wind is an analytic height-based deformation, not botanical physics: it does not preserve exact branch lengths or simulate inertia/collisions. Nearby foliage can overlap.

- Gardens are session-only; refresh resets them. Desktop WebGL2 is required; mobile/touch is outside the approved scope.

- Movable-light bending, additional species, and shadows remain scheduled work. Only the current five-plant workload is validated, not the original 15–20-plant target.

- Results are from Chrome on one Apple M2 GPU. Cross-browser/GPU coverage and shader-startup/performance headroom need further testing.

Validation and measured performance

Seven unit tests and six Chrome browser tests pass. Checks cover staged growth, seed replay, five-plant bounds, the full session loop, pause/resume, moving light anchors, daylight controls, repeated reset/resource bounds, camera drags, diagnostic cancellation, and WebGL2 failure.

1 plants: 59.5 FPS / 16.8 ms p95 / 23 calls, 3 plants: 60.0 FPS / 16.7 ms p95 / 33 calls, 5 plants: 60.0 FPS / 16.7 ms p95 / 43 calls.

Measured on Apple M2, Chrome 156.0.0.0, 1086 x 594 canvas, DPR 1; wind 70%, daylight 15%, bloom control 0.65. Each level: 10s growth/warm-up + 8s measured RAF intervals. Slowest recorded frame: 83.4 ms. Averages near 60 FPS do not establish headroom or performance on other GPUs. These are frame intervals, not GPU timings. Raw evidence: docs/alpha/benchmark-results.json.